

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4_AT2 — VISUALIZATION EVALUATION EXERCISE

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO4 – Evaluation (BL5)	Assessment:	Assessment Tool 2 – Visualization Evaluation Exercise
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO4 Statement:	<i>Evaluate the appropriateness and design choices of relationship-visualization techniques (scatterplots, correlograms, PCA, bubble/hexbin plots, and network graphs) and justify improvements.(BL5)</i>		
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Section / Batch:	2024	Date:	22-08-26

Code of Conduct

I Nivya Sri T (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nivya Sri T

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond describing the scenario: evaluate the specific strength or weakness, and justify your judgment.
- Where a question asks for a recommendation, propose a specific alternative — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION EVALUATION EXERCISE

Rubrics for Calculation **ASSESSMENT RUBRIC**
AT2 — Visualization Evaluation Exercise

CO4 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO4-AT2 Visualization Evaluation Exercise problems, in which students are given a visualization design choice and must evaluate its appropriateness, identify limitations, and justify improvements.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant visualization principle being evaluated (e.g., overplotting, PCA variance, encoding choice, network readability).	Good understanding of the relevant principle, with only minor gaps.	Basic understanding, but with some misconceptions about which principle applies.	Poor understanding of core principles; misidentifies the nature of the issue entirely.
Explanation	25%	Clearly explains why the design choice is effective or ineffective, including its practical consequence for a reader — not just that it is 'good' or 'bad.'	Explains the evaluation with adequate reasoning, covering most of the practical consequence.	Identifies the issue but explanation is generic or only loosely tied to its practical effect.	Little or no explanation provided; the evaluation is a bare assertion.
Application	20%	Proposes a specific, well-justified recommendation or alternative that directly addresses the evaluated issue.	Proposes a workable recommendation, with only minor gaps in justification.	Proposes a recommendation, but it only partially addresses the issue or lacks justification.	Fails to propose a workable recommendation, or it does not address the evaluated issue.
Organization	15%	The response is clearly structured, addressing evaluation and recommendation in a logical sequence with coherent overall flow.	The response is organized, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the evaluation and	Mostly accurate, with only minor	Partially correct; the evaluation or	Largely incorrect or inappropriate,

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		the recommendation are completely accurate and well-suited to the specific scenario described.	omissions or a small error.	recommendation contains notable inaccuracies.	with major gaps in both the evaluation and recommendation.

Scoring Guide

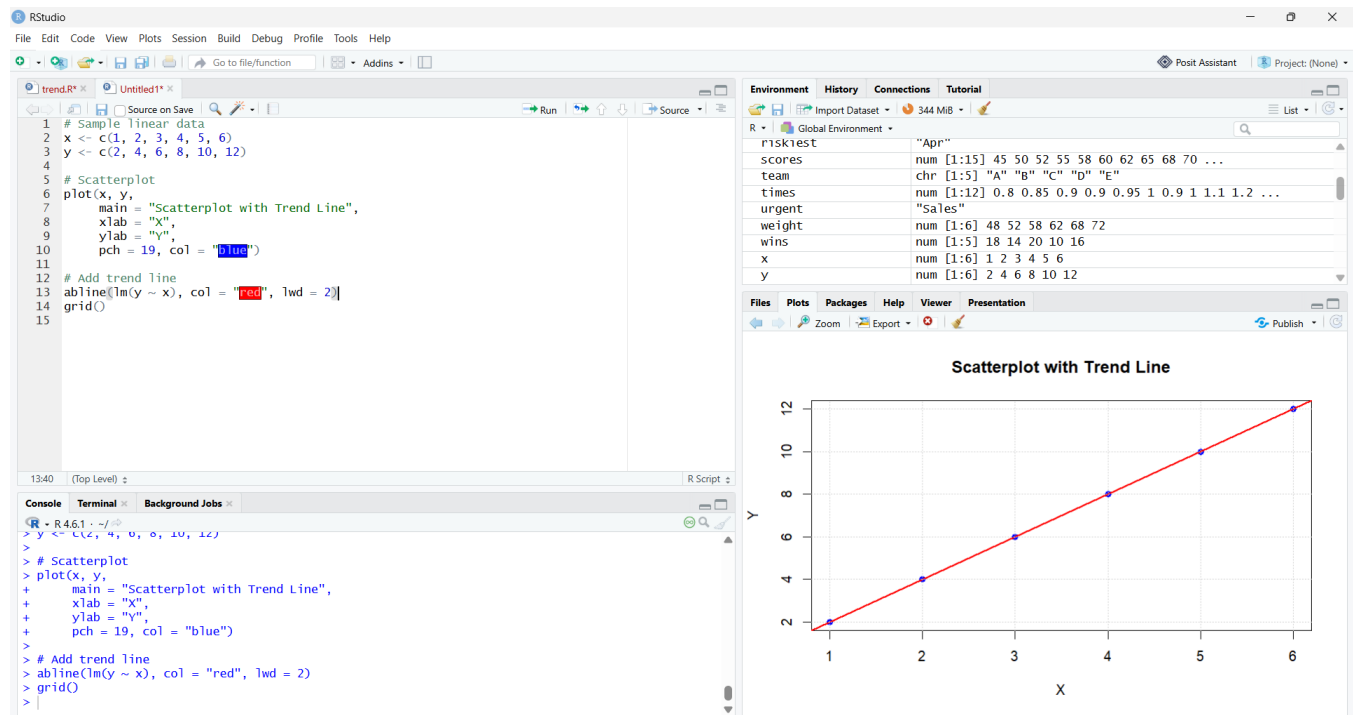
For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

PROBLEM 3:

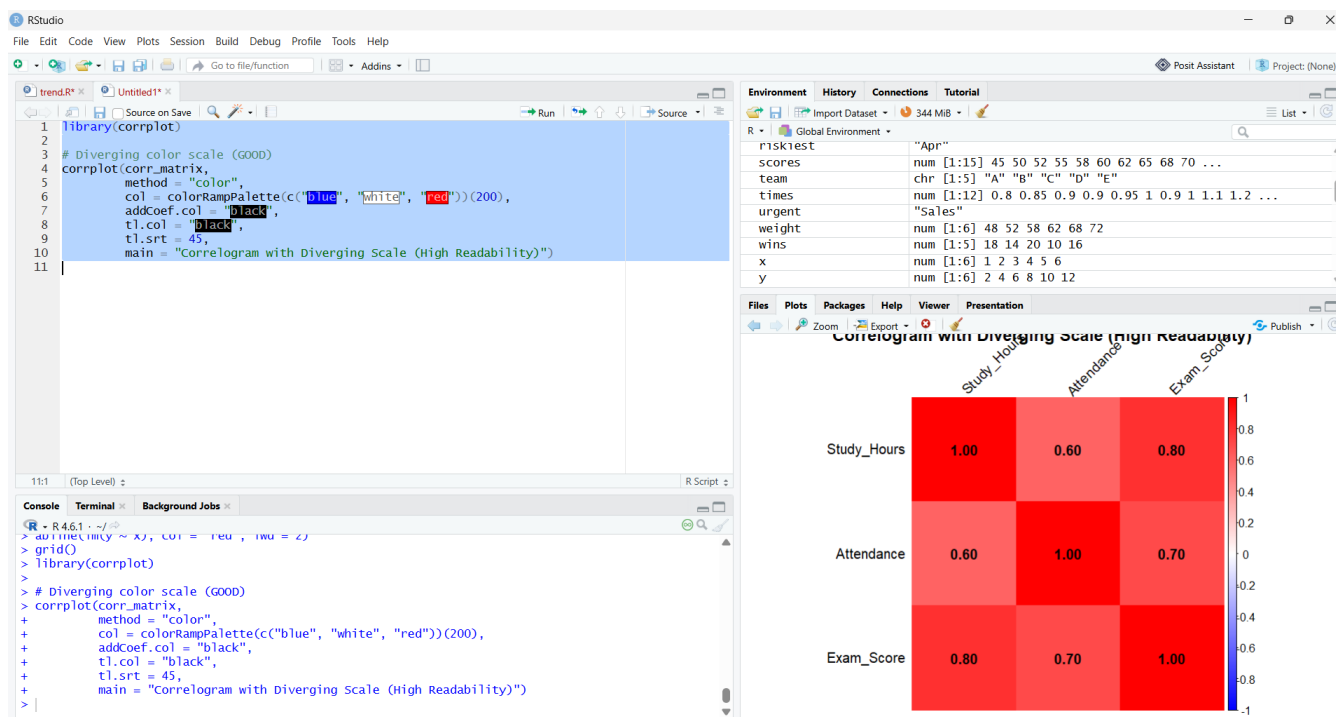
- The scatterplot alone shows points but does not clearly communicate the linear pattern.
- Adding `abline(lm(y ~ x))` makes the trend explicit, improving:
 - Interpretation
 - Clarity
 - Analytical value

This demonstrates why omitting the trend line reduces usefulness.



PROBLEM 7:

- The rainbow version shows abrupt color jumps → misleading visual boundaries.
- The diverging version shows a smooth transition from negative → zero → positive correlations.
- This directly demonstrates why rainbow palettes reduce readability and distort interpretation.



PROBLEM 11:

1. PCA is scale-dependent

PCA uses variance, and variance depends on measurement units. Large-scale variables (income in dollars) have huge variance compared to small-scale variables (age in years).

2. High-scale variables dominate PC1

Without standardization:

- PC1 becomes almost entirely driven by income
- Age is nearly ignored
- PCA results become biased and misleading

3. Distorted loadings and interpretations

Loadings appear large for income and tiny for age only because of scale, not importance. This leads to wrong conclusions about which variables matter.

4. PCA no longer reflects true data structure

The principal components fail to capture real relationships. The analysis becomes unreliable.

Not standardizing variables before PCA creates a major risk: the principal components become dominated by high-scale variables, producing biased, misleading results that do not represent the true patterns in the dataset.

The screenshot shows the RStudio interface. The source editor contains the following R code:

```
1 # Data: income (dollars) vs age (years)
2 income <- c(40000, 50000, 60000, 80000, 120000)
3 age <- c(25, 30, 35, 40, 45)
4
5 data <- data.frame(income, age)
6
7 # PCA with scaling (GOOD)
8 pca_scaled <- prcomp(data, scale. = TRUE)
9 summary(pca_scaled)
10 pca_scaled$rotation
11
12
```

The console shows the execution of the code:

```
R - R 4.6.1 - ~/
> # PCA with scaling (GOOD)
> pca_scaled <- prcomp(data, scale. = TRUE)
> summary(pca_scaled)
Importance of components:
              PC1      PC2
Standard deviation   1.396 0.2236
Proportion of Variance 0.975 0.0250
Cumulative Proportion 0.975 1.0000
> pca_scaled$rotation
              PC1      PC2
income 0.7071068 0.7071068
age     0.7071068 -0.7071068
>
```

PROBLEM 20:

1. Why This Is a Mistake (Conceptual Issue)

Bubble charts encode magnitude using **area**, not radius. But area grows with the **square of the radius**:

$$\text{Area} = \pi r^2$$

So if radius doubles, area becomes **four times larger**.

This means:

- A bubble sized by radius exaggerates differences
- The largest bubble becomes visually dominant
- The viewer perceives a much bigger gap than the data justifies

2. Visual Effect on the Chart

a) Over-emphasis of large values

The biggest bubble appears **massively inflated**, drawing attention away from other data points.

b) Distorted comparisons

Differences between values look **non-linear** and **overstated**. A value that is only 2× larger may appear 4× or 5× larger visually.

c) Misleading interpretation

Readers may conclude:

- One category is overwhelmingly dominant
- Smaller categories are insignificant

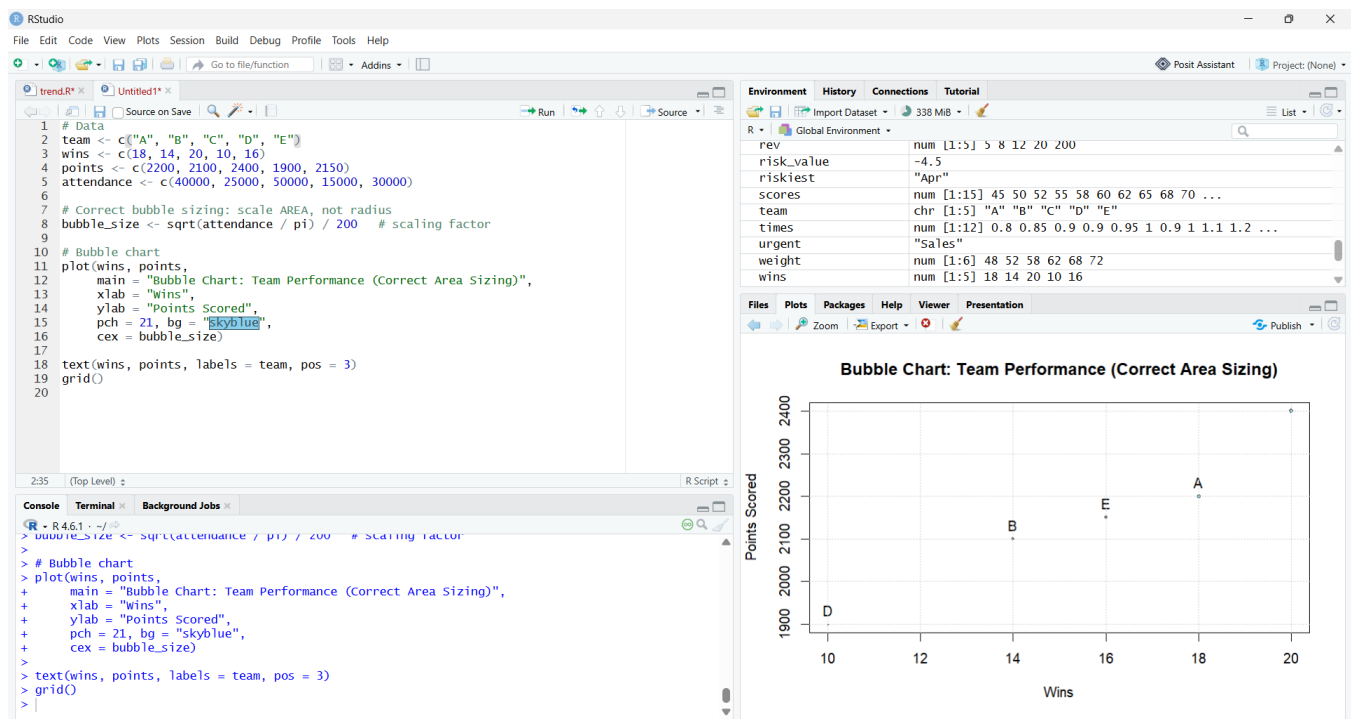
Even when the actual data differences are moderate.

3. Impact on Data Integrity

This mistake breaks the fundamental rule of quantitative visualization: **Visual encoding must match the underlying numeric scale**.

Using radius violates this rule and produces:

- Biased perception
- Incorrect conclusions
- Reduced trust in the chart



PROBLEM 24:

A **network graph** is not appropriate for a simple linear process like **Step 1 → Step 2 → Step 3 → Step 4**. It adds unnecessary complexity and reduces clarity compared to simpler sequential charts.

Evaluation of the Chart Choice

1. Network graphs are meant for complex relationships

Network graphs are designed for:

- Multiple connections
- Branching
- Loops
- Clusters
- Many-to-many relationships

Your process has **none** of these. It is a **single straight path**, not a network.

Using a network graph here misrepresents the structure.

2. Visual clutter without added insight

A network graph introduces:

- Nodes
- Edges
- Spatial layout algorithms

These elements **do not improve understanding** of a simple linear sequence. Instead, they make the viewer work harder to interpret something that should be obvious.

3. Better alternatives exist

For a linear process, clearer chart types include:

- **Flowchart**
- **Process diagram**
- **Timeline**
- **Ordered list**
- **Arrow diagram**

These formats communicate sequence **directly and intuitively**.

4. Risk of misinterpretation

A network graph may incorrectly suggest:

- Branching
- Parallel steps
- Dependencies
- Multiple pathways

This can confuse readers and distort the intended message.